

Advanced Encryption Schemes

Klevis Imeri Lecture Notes

Autumn 2026

Contents

1	Lecture 1	1
2	Lecture 2	2

1 Lecture 1

Definition 1.1 (Probabilistic polynomial-time algorithm).

$$\text{PPT}(A) \iff \exists p \in \text{poly} \quad \forall x \in \{0, 1\}^* \quad \forall r : \text{time}_A(x; r) \leq p(|x|).$$

Definition 1.2 (Negligible function).

$$\mu : \mathbb{N} \rightarrow \mathbb{R} \text{ negligible} \iff \forall c \in \mathbb{N}_{>0} \exists \lambda_0 \in \mathbb{N} \forall \lambda \geq \lambda_0 : |\mu(\lambda)| < \frac{1}{\lambda^c}.$$

Definition 1.3 (Public key encryption scheme).

$$\Pi := (\text{Gen}, \text{Enc}, \text{Dec}).$$

$$\text{PKE}(\Pi) \iff \left\{ \begin{array}{l} \text{PPT}(\text{Gen}) \wedge \text{PPT}(\text{Enc}) \wedge \text{PPT}(\text{Dec}), \\ \forall \lambda \in \mathbb{N}_{>0} : \\ \quad \text{Gen}(1^\lambda) \in \mathcal{K}_{\text{pub}, \lambda} \times \mathcal{K}_{\text{sec}, \lambda}, \\ \quad \forall pk \in \mathcal{K}_{\text{pub}, \lambda} \quad \forall m \in \mathcal{M}_\lambda : \\ \quad \quad \text{Enc}(pk, m) \in \mathcal{C}_\lambda, \\ \quad \forall sk \in \mathcal{K}_{\text{sec}, \lambda} \quad \forall ct \in \mathcal{C}_\lambda : \\ \quad \quad \text{Dec}(sk, ct) \in \mathcal{M}_\lambda \sqcup \{\perp\}. \end{array} \right.$$

$\text{Gen}, \text{Enc}, \text{Dec}$: algorithms,

λ : security parameter,

$\mathcal{K}_{\text{pub}, \lambda}, \mathcal{K}_{\text{sec}, \lambda}$: public and secret key spaces,

$\mathcal{M}_\lambda, \mathcal{C}_\lambda$: message and ciphertext spaces,

$\perp$: decryption failure, $\sqcup$: disjoint union.

Definition 1.4 (Perfect correctness).

$$\text{PerfectCorrect}(\Pi) \iff \forall \lambda \in \mathbb{N}_{>0} \quad \forall (pk, sk) \in \text{Gen}(1^\lambda) \quad \forall m \in \mathcal{M}_\lambda : \\ \Pr[\text{Dec}(sk, \text{Enc}(pk, m)) = m] = 1.$$

2 Lecture 2